



## **BIF401 LIVE (QUIZ)**

### **MID TERM**

**Past papers for all subjects are also  
available 03184148783**

1. Which one of the following statements does not apply to BLAST Algorithm?
  - It is one of the tools of NCBI
  - It stands for Basic Local Alignment Search Tool
  - Even if no word/base are similar, there is an alignment to be considered
  - It uses word matching like FASTA algorithm
  - **Correct answer: Even if no word/base are similar, there is an alignment to be considered ✓**
2. Compares protein sequence against protein databases.
  - tblastx
  - blastn
  - blastp
  - blastx
  - **Correct answer: blastp ✓**
3. What happens when we process so many progressive alignments in a single alignment process?
  - **Alignment becomes slow ✓**
  - Alignment becomes fast
  - None of these
  - No effect on process speed
4. The expansion of BLAST is:



- Basic Local Alignment Search Tool ✓
  - Bioinformatics Logical Alignment Search Tool
  - Basic Logical Alignment Search Tool
  - Beginners Local Alignment Search Tool
5. **Fastm35 compares ..... to a protein sequence database.**
- DNA
  - ordered peptides
  - un-ordered peptides
  - Proteins ✓
1. **Which of the following statements about amino acids is correct?**
- Amino acids are classified according to the structures and properties of their side chain ✓
  - Amino acids in proteins are mainly in the D-configuration
  - Amino acids are un-charged at neutral pH
  - Twenty-four amino acids are commonly used in protein synthesis
2. **Analysis of a sample of DNA found that 20% of the bases were Adenine (A). What percentage of the bases would be Pyrimidine?**
- 50% ✓
  - 60%
  - 30%
3. **What is the real challenge in bioinformatics?**
- To analyze protein structures
  - To make sense of biological data ✓
  - To identify new genes
  - To sequence genomes
4. **What term is used to describe the process by which a segment of DNA is copied to produce a molecule of messenger RNA?**
- Translation



- Transcription ✓
  - Replication
  - Reproduction
5. Which instruments/technologies are used for obtaining digitalized biological information of genes and proteins?
- X-ray Machine
  - Next Generation Sequencing (NGS) and High Resolution Mass Spectrometry ✓
  - High Resolution Mass Spectrometry
  - Next Generation Sequencing (NGS)
1. What term is used to describe the process by which a segment of DNA is copied to produce a molecule of messenger RNA?
- Transcription ✓
  - Reproduction
  - Replication
  - Translation
2. What is "Bioinformatics"?
- A software for analyzing digital images of biological samples
  - A method for growing cells in a lab
  - An experimental technique for studying biological systems
  - An interdisciplinary field involving biology, mathematics, computer science, chemistry and physics ✓
3. Which of the following statements about amino acids is correct?
- Amino acids are classified according to the structures and properties of their side chain ✓
  - Amino acids are un-charged at neutral pH
  - Amino acids in proteins are mainly in the D-configuration
  - Twenty-four amino acids are commonly used in protein synthesis
4. A nucleotide is formed of which of the following units?



- Nitrogen base, sugar and phosphate ✓
- Nitrogen base and sugar
- Nitrogen base and phosphate
- Sugar and phosphate

5. Statement "Dot plots cannot capture insertions, deletions and gap indications".

- True Statement
- False Statement ✓

. Which of the following options, A — D, are the Pyrimidine bases found in DNA?

- Uracil & Cytosine
- Cytosine & Thymine ✓
- Adenine & Thymine
- Thymine & Uracil

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2. The methodology/ procedure of aligning two sequences by searching for patterns that is in the same order in the sequences is known as

- Multiple Sequence Alignment
- Sequence Alignment
- All of these
- Pairwise Alignment ✓

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3. What is "Bioinformatics"?

- A software for analyzing digital images of biological samples
- A method for growing cells in a lab
- An interdisciplinary field involving biology, mathematics, computer science, chemistry and physics ✓
- An experimental technique for studying biological systems



Which is affected by extreme values?

Mode

Median

Mean ✓

All of these

"An aggregate of entities for which we have an interest at a particular time" is called:

Population ✓

Mean

Sample

Which information is not correct about sample?

A sample is typically only a part of the whole population

Sample data provide only limited information about the population

None of these ✓

Sample statistics are generally imperfect representatives of the corresponding population parameters

An assumption about the population parameter is:

Hypothesis ✓

Inference

Probability

Significance

Which one is not the type of Probability Sampling designs?

Non-systematic sampling ✓

Systematic sampling

Cluster (area) sampling

Multistage sampling

Sometimes ----- is also called as "proportional or "quota" random sampling.

Simple sampling

Cluster sampling

Systematic sampling

Stratified sampling ✓

Type of sampling refers to identifying someone who meets the criteria for inclusion in the study.

Probability sampling

Random sampling

Non Probability sampling ✓

Snowball sampling

Which case is true for simple random sampling?

Can sample with or without replacement ✓

Can sample with replacement

Can sample without replacement



**A discrete variable is:**

Skull circumference

Height

**The number of daily admissions in a hospital ✓**

**Qualitative Variables include:**

Weight of students

Height

Hemoglobin values

**Blood group ✓**

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